Bureau of Indian Standards
Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 383:2016

2016

COARSE AND FINE AGGREGATES FOR CONCRETE

SPECIFICATION

ICS 91.100.15
BIS 2016

Indian Standard

1 SCOPE

This standard covers the requirements for aggregates obtained from natural sources for use in concrete. It specifies grading requirements, physical properties, and chemical limits for both coarse and fine aggregates.

2 CLASSIFICATION

2.1 Fine Aggregate (Sand)

Fine aggregate is material passing through 4.75 mm IS sieve and retained on 150 micron IS sieve. It is classified as Zone I, Zone II, Zone III, or Zone IV based on particle size distribution.

2.2 Coarse Aggregate

Coarse aggregate is material retained on 4.75 mm IS sieve. It is classified as single sized or graded aggregate with nominal sizes of 10mm, 12.5mm, 16mm, 20mm, 25mm, 31.5mm, and 40mm.

3 GRADING REQUIREMENTS - FINE AGGREGATE

IS Sieve Size	Zone I	Zone II	Zone III	Zone IV
4.75 mm	100	100	100	100
2.36 mm	90-100	90-100	90-100	90-100
1.18 mm	60-95	75-100	85-100	95-100
600 m	35-59	60-84	75-100	90-100
300 m	8-30	15-44	25-59	40-79
150 m	0-10	0-16	2-25	10-35

4 PHYSICAL PROPERTIES

Property	Fine Aggregate	Coarse Aggregate
Specific Gravity	2.5 - 3.0	2.5 - 3.0
Water Absorption, %	2.0 max	2.0 max
Aggregate Impact Value, %	-	30 max (for non-wearing)
Aggregate Crushing Value, %	-	30 max
Abrasion Value, %	-	30 max (Los Angeles)
Fineness Modulus	2.0-3.5	-

5 CHEMICAL LIMITS

- Chloride Content: 0.05% by mass of aggregate (for prestressed concrete)
- Sulphate (as SO3): 0.5% by mass of aggregate
- Alkali Reactivity: Mortar bar expansion 0.05% at 12 months
- Organic Impurities: Color of solution shall not be darker than reference